Introduction to Robotics

Robotics is a branch of AI that deals with designing, building, and using robots. A **robot** is a machine that can perform tasks automatically, often by mimicking human actions.

Main Parts of a Robot

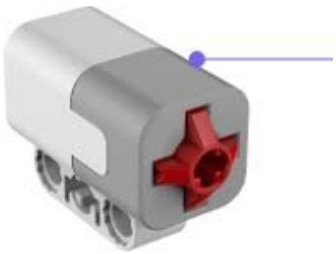
- **Sensors – "The robot's eyes and ears"**

Sensors help a robot detect and understand its surroundings.

Examples of Sensors:

- **Ultrasonic Sensor** – Detects distance (like a bat using sound)
- **Infrared Sensor (IR)** – Detects heat or objects
- **Camera** – For seeing and recognizing objects
- **Touch Sensor** – Feels pressure or touch

Example Touch Sensor



- **Actuators – "The robot's muscles"**

Actuators are the parts that help a robot move or perform actions.

Examples of Actuators:

- **DC Motors** – Used for rotating wheels
- **Servo Motors** – For precise movements like arm control



- **Controllers – "The robot’s brain"**

Controllers process information from sensors and tell actuators what to do.

Examples of Controllers:

- **RCU M6** – A compact robot control unit.
- **Arduino** – Popular in DIY and hobby robots
- **Raspberry Pi** – A mini-computer used for smarter robots

Example RCU



- **Beams and Connectors –** ***"The robot’s skeleton and joints"***

These parts give the robot its **shape** and **structure**.

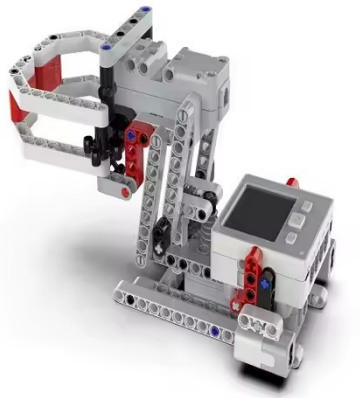
- **Beams:** Long pieces used to build the robot’s frame or arms
- **Connectors:** Join parts together .



Example of a Robot with Actuators - Motors, Controller – RCU, Sensors, Beams and Connectors









Practicals